



Latitudes — FULL MIND MAP

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Chapter 02_i_Latitudes

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Latitude Basics — Range aur Great Circles

- Valid latitude range: **90° North se 90° South** tak hi possible hai (91° North jaisi values invalid); 91° East/West **longitude** ho sakti hai, latitude nahi. [Q1]
- Great Circle = wo circle jo globe ko 2 equal hemispheres mein divide kare — sirf **Equator** (latitudes mein) aur har meridian+opposite meridian pair great circle banate hain. Baaki koi bhi latitude (jaise 60° N) great circle **nahi** hoti. [Q2]

⚠ EXAM TRAP

- Sirf **Equator** hi ek latitude hai jo great circle hai — baaki sab parallels chhote hote hain. [Q2]

0° Latitude x 0° Longitude Intersection

- Equator (0° lat) aur Prime Meridian (0° long) ka intersection point **Atlantic Ocean** mein hai (Gulf of Guinea). [Q3,Q4]

Coordinate se Location Identify Karna

- $\sim 5^\circ$ E, 50° N coordinate **Belgium (Europe)** ke close match karta hai. [Q5]
- Antipode calculation: latitude sign flip hota hai (30° N $\rightarrow$ 30° S); longitude ke liye 180° se subtract karke direction flip karo (50° E $\rightarrow$ $180-50=130^\circ$ W). [Q6]

⚠ EXAM TRAP

- Antipode longitude nikalne ke liye simple opposite mat lo — formula hai: $180^\circ - \text{given longitude}$, phir direction (E $\leftrightarrow$ W) flip karo. [Q6]

Cities — Latitude Comparison

- Given cities mein **Rome ($41^\circ 54'$ N)** sabse north hai (Beijing $39^\circ 55'$ N, New York $40^\circ 45'$ N, New Delhi $28^\circ 38'$ N se zyada). [Q7]

- Sabse zyada north-south (latitudinal) extension wale desh ka answer 'More than one' hai — Chile ka continuous mainland extent sabse zyada hai, lekin Russia ka total extent (islands including) usse bhi zyada hai. [Q8]

1° Interval Latitude Lines — Counting

- Agar 1° interval pe latitude lines banayi jaayein: Northern Hemisphere mein 89 + Southern Hemisphere mein 89 + Equator 1 = **total 179** lines. [Q9]